

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
Answer Key <i>Scheme of evaluation (marks in brackets) and answers of problems/key</i>			
FOURTH SEMESTER B.TECH DEGREE EXAMINATION July 2021 (2019 Scheme)			
Course Code: HUT200			
Course Name: Professional Ethics			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions. Each question carries 3 marks	Marks
	1	What are the two approaches to Engineering ethics? There are conventionally two approaches in the study of ethics: 1. Micro-ethics which deals with decisions and problems of individuals, professionals, and companies. 2. Macro-ethics which deals with the societal problems on a regional/national level. For example, global issues, collective responsibilities of groups such as professional societies and consumer groups.	(3)
	2	List two methods of developing self confidence. The following methodologies are effective in developing self-confidence in a person: 1. Encouraging SWOT analysis. By evaluating their strength and weakness, they can anticipate and be prepared to face the results. 2. Training to evaluate risks and face them (self-acceptance). 3. Self-talk . It is conditioning the mind for preparing the self to act, without any doubt on his capabilities. This make one accepts himself while still striving improvement. 4. Study and group discussion, on the history of leaders and innovators (e.g., Sam Walton of Wal-Mart, USA).	(3)
	3	What are the situations when moral dilemmas arise? There are some difficulties in arriving at the solution to the problems, in dilemma. The three complex situations leading to moral dilemmas are: 1. The problem of <i>vagueness</i>: One is unable to distinguish between good and bad (right or wrong) principle. Good means an action that is obligatory. For example, code of ethics specifies that one should obey the laws and follow standards. Refuse bribe or accept the gift, and maintain confidentiality 2. The problem of <i>conflicting reasons</i>: One is unable to choose between two good moral solutions. One has to fix priority, through knowledge or value system. 3. The problem of <i>disagreement</i>: There may be two or more solutions and none of them mandatory. These solutions may be better or worse in some respects but not in all aspects. One has to interpret, apply different morally reasons, and analyze and rank the decisions. Select the best suitable, under the existing and the most probable conditions.	(3)

4	What are the types of ethics depending upon the morality of humanity? Consequential Ethics The Consequential ethics are values the outcome of which determine the morality behind a particular action. A lie which saves a life, comes under this. Non-consequential Ethics The non-consequential ethics are values followed where the source of morality comes from the standard values. The moral law which states that a lie is a lie, and shouldn't be done, though it ends in a good deed can be taken as an example of nonconsequential ethics.	(3)												
5	Give three conditions, essential for valid informed consent. Consent is voluntary, all relevant information shall be presented in a clearly understandable form, and consenters shall be capable of processing information and make rational decisions.	(3)												
6	What are the advantages of codes of ethics? Serving and protecting the public – Engineers are in a responsible position where trust and trustworthiness, both are essential. Guidance – Codes are written in brief yet prove effective in offering general guidance to the engineers. Inspiration – Codes of ethics, which specify a collective commitment towards a profession, help in motivating the engineers towards ethical conduct. Shared Standards – The standards established should be applicable to all individuals, in their particular professions.	(3)												
7	Differentiate between copyright and trademark. <table><thead><tr><th>Trademark</th><th>Copyright</th></tr></thead><tbody><tr><td>A trademark protects items that help define a company brand</td><td>Copyright is geared toward literary and artistic works</td></tr><tr><td>It protects enterprise, logo, product names, phrases, slogans etc.</td><td>It protects books, articles, web contents, paintings, photographs, songs, movies, recorded shows, choreography etc.</td></tr><tr><td>Trademark manifests exclusive rights to the owner of trademark to use it to assign license to another party to use the same</td><td>It is a bundle of rights conferred by law that protects the original piece of work from being copied</td></tr><tr><td>It identifies the brand and separates it from other products in the market.</td><td>It prevents others from publishing or copying the original piece of work.</td></tr><tr><td>Issued for comparatively shorter period.</td><td>Issued for long term.</td></tr></tbody></table>	Trademark	Copyright	A trademark protects items that help define a company brand	Copyright is geared toward literary and artistic works	It protects enterprise, logo, product names, phrases, slogans etc.	It protects books, articles, web contents, paintings, photographs, songs, movies, recorded shows, choreography etc.	Trademark manifests exclusive rights to the owner of trademark to use it to assign license to another party to use the same	It is a bundle of rights conferred by law that protects the original piece of work from being copied	It identifies the brand and separates it from other products in the market.	It prevents others from publishing or copying the original piece of work.	Issued for comparatively shorter period.	Issued for long term.	(3)
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8	What is meant by ‘Occupational Crime’? Wrong actions of a person through one’s lawful employment, Crime committed by employee to promote his interest, and theft by the employee	(3)												
9	List any three characteristics of Business Ethics. 1. A business should aim to have fair dealing with everyone dealing with it.	(3)												

		2. Ethics should be fixed for everyone working in the organisation at any level and their implementation should be linked with reward-punishment system. 3. Any violation of ethics should be detected at the earliest and remedial measures taken immediately. 4. Business ethics should be based on broad guidelines of what should be done and what should be avoided. 5. The ethics should be based on the perception of what is right.	
	10	List any three ethical responsibilities of consulting engineer. 1. Advertising The consulting engineers are directly responsible for advertising their services, even if they employ other consultants to assist them. But in many organizations, this responsibility is with the advertising executives and the personnel department. 2. Competitive Bidding It means offering a price, and get something in return for the service offered. The organizations have a pool of engineers. 3. Contingency Fee This is the fee or commission paid to the consultant, when one is successful in saving the expenses for the client.	(3)
PART B			
<i>Answer any one full question from each module. Each question carries 14 marks</i>			
Module 1			
	11	Explain about academic integrity and write the five pillars of academic integrity. <i>The International Center for Academic Integrity (ICAI) has identified five pillars of academic integrity to generate positive conversations about integrity.</i> Pillar 1 Honesty is sincerity. All other pillars of academic integrity have some basis in honesty. Honest individuals take stock of individual abilities and represent their effort fairly. Pillar 2 Trust in other people and in your community eases working relationships. Trust is established in a system where all members are doing their best work, where structures and policies are fair and all will be treated fairly. Pillar 3 Fairness goes hand in hand with trust. Every individual should believe that they will be treated fairly and judged by the same standard as all others in the community. Pillar 4 Respect allows for individual points of view and opinions to be shared. Students show respect by “listening to other points of view, being prepared, meeting deadlines, and performing to the best of	(14)

		their ability.” Instructors show respect by listening to students’ ideas and “providing full and honest feedback.” Pillar 5 Responsibility means acknowledging your agency and accountability in daily actions and in your work. Everyone is personally invested in performing their work with integrity and encourages others to act with integrity too.	
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	12	Explain the core elements of a strong work ethics. Honesty , Integrity, Impartiality/ Fairness, Alertness, Openness, Respect for others, Reliability and dependability, Determination, Dedication, Accountability, Confidentiality, Responsibility, Legality, Competence, Professionalism, Humility, Initiative) - Any 14 points with short explanation .	(14)
Module 2			
	13(a)	a) Explain the three main levels of moral developments, devised by Carol Gilligan. Carol Gilligan opines that Kohlberg’s theories are biased upon the male thinking process. According to Gilligan, Kohlberg seemed to have studied only privileged men and boys. She believed that women face a lot of psychological challenges and they are not moral widgets. The women’s point of view on moral development involves caring which shows its effect on human relationships. Pre-conventional Level ☐ A person in this stage cares for oneself to ensure survival. ☐ Though the person’s attitude is selfish, this is the transition phase, where the person finds the connection between oneself and others. Conventional Level ☐ In this stage, the person feels responsible and shows care towards other people. ☐ Carol Gilligan believes that this moral thinking can be identified in the role of a mother and a wife. This sometimes leads to the ignorance of the self. Post-conventional Level ☐ This is the stage, where the principle of care for self as well as others, is accepted. ☐ However, a section of people may never reach this level.	(7)
	13(b)	Discuss on three types of inquiries. Normative Inquiries These inquiries are mostly helpful to identify the values which guide the individuals and groups in taking a decision. These are meant for identifying and justifying some norms and standards of morally desirable nature for guiding individuals as well as groups. Conceptual Inquiries These are meant for describing the meaning of concepts, principles, and issues related to Engineering Ethics. These inquiries also explain whether the concepts and ideas are expressed by single word or by phrases.	(7)

	14	List and explain the varieties of moral issues 1. Organization oriented issues • Being an employee to firm, the engineer has to work towards the achievement of the objectives of his/her organization. • Engineers have to give higher priority to the benefits of the organization than one's own benefits. 2. Clients or customers oriented issues • As we know, the purpose of any business is to reach and satisfy the end users. Therefore the customers' requirements should be met. 3. Competitors oriented issues • In order to withstand in a market, engineers should produce things better than their competitors by all means. • But engineers should not practice cut-throat competition. They should follow certain professional behavior while facing their competitors. 4. Law, government and public agencies oriented issues • The engineers should obey and voluntarily comply with all the governmental rules and regulations related to them. 5. Professional societies oriented issues • The engineers should follow strictly the various codes of ethics by various professional societies such as National Society of Professional Engineers (NSPE), the Institute of Electrical and Electronics Engineers (IEEE), and American Society of Mechanical Engineers (ASME), in order to perform standard professional behavior. 6. Social and environmental oriented issues • Since the works of engineers have a direct and vital impact on the quality of life for all people, the engineers should be dedicated to the protection of the public health, safety and welfare.	(14)
Module 3			
	15	Explain about Bhopal Gas Tragedy and write its cause and fatal effect. Bhopal's Gas tragedy is the world's worst industrial disaster that occurred in 1984, due to the gas leakage from a pesticide production plant, The Union Carbide India, Limited (UCIL) located in Bhopal, Madhya Pradesh. It was believed that slack management and deferred maintenance together created a situation where routine pipe maintenance caused a backflow of water into the MIC tank, triggering the disaster. In the early hours of December 3rd, 1984, a rolling wind carried a poisonous gray cloud from the Union Carbide Plant in Bhopal, Madhya Pradesh of India. The poisonous gas released was 40 tons of Methyl Iso Cyanate (MIC). This particular gas is very toxic that leaked and spread throughout the city. The residents of the city, woke up to the clouds of suffocating gas and struggled to breathe. They started running desperately through the dark streets. The victims arrived at hospitals, breathless and blind.	(14)

	The people who survived had their lungs, brain, eyes, muscles affected severely. Their gastro intestinal system, neurological, reproductive and immune systems were also dangerously affected. By the morning, when the sun rose clearly, the roads were all filled with dead bodies of humans and animals, the trees turned black and the air filled with foul smell. The Union Carbide Corporation (UCC) team and also the CBI (Central Bureau of Investigation) team conducted separate investigations on the cause of the incident and came to the same conclusion. It was understood that a large volume of water had been released into the MIC tank and this further caused a chemical reaction that forced the pressure release valve to open and allowed the gas to leak. Initial effects of exposure were – • Coughing • Severe eye irritation • Feeling of suffocation • Burning sensation in the respiratory tract • Blepharospasm • Breathlessness • Stomach pains • Vomiting	
16	Explain the Babylon's Building Code and The United States Steamboat Code. Babylon's Building Code – 7 marks This code was set by Hammurabi, king of Babylon. It aimed at the builders of his time wherein, they were forced to follow the code by law. He ordered them, “If a builder has built a house for a man and has not made his work sound, and the house which he has built was fallen down and so caused the death of the householder, that builder shall be put to death. If it causes the death of the house holder's son, they shall put that builder's son to death. If it causes the death of the house holder's slave, he shall give slave to the householder. If it destroys property, he shall replace anything it has destroyed; and because he has not made the house sound which he has built and it has fallen down, he shall rebuild the house which has fallen down from his own property. If a builder has built a house for a man and does not make his work perfect and the wall bulges, that builder shall put that wall into sound condition at his own cost”. The Unites States Steamboat Code – 7 marks The steam engines used for travel during those days were really heavy and bulky. James Watt who invented steam engine worked with two more scientists Oliver Evans and Richard Trevithick who had modified the old steam engines by removing condensers and made them compact. These redesigned engines though made lighter, couldn't solve the problem of boiler explosions. The speed of the boats if increased led to the explosion of the boilers on steam boats causing disasters. Then Alfred	(14)

	Guthrie, an engineer of Illinois had inspected around 200 steam boats with his own funding and found out the reasons for the boiler explosions and later prepared a report relating to the care that could be taken later. The recommendations made by him were published by Senator Shields of Illinois and incorporated in senate documents which later was made a law, which made the mechanical engineers of America (ASME), to formulate the standards in the manufacturing of steam boats.	
Module 4		
17(a)	Explain the methods for managing conflict. Conflict in any work environment is inevitable for the simple reason that whenever people have to work together they will not always be in perfect agreement on all issues, goals, or perceptions. From this emerges a definition of conflict – a “disagreement between two or more organisational members or groups arising from the fact they must share resources of work activities and/or from the fact that they have different status, goals, values or perceptions.” In short, conflict refers to any kind of opposition or antagonistic intersection between two or more parties. Collective bargaining – 2 marks Conciliation and mediation – 2 marks Arbitration – 3 marks	(7)
17(b)	Explain the types of Collective Bargaining. There are four main types of collective bargaining – • Distributive Bargaining – In this, one party’s gain is another party’s loss. Example – Wages • Integrative bargaining – In this, both the parties may gain or none of the parties may face a loss. Example – Better training programs • Attitudinal Structuring – When there is backlog of bitterness between both the parties then attitudinal structuring is required to make smooth industrial relations. • Intra-organizational Bargaining – There can be conflicting groups in both management and unions also. So, there is need to achieve consensus in these groups.	(7)
18	Explain the steps taken for conflict management When a conflict, arises whether intra-individual, inter- individual; intra-group or inter- group in an organisation, it must be resolved as early as possible. In an organisation, there must be someone to intervene before the situation goes worse and generally a superior helps to improve the situation. In order to resolve the conflict effectively the seniors or superior should handle the situation carefully and take the following steps:	(14)

	1. Preliminary step — knowing the conflict; 2. Diagnosing the issue; 3. Applying any of the conflict handling modes. 1. Preliminary Step: The first stage in resolving the conflict is to know the full details of the conflict. As soon as the conflict comes to the knowledge of the senior or superior, he should handle the conflict skillfully. 2. Diagnosing the Issue: In diagnosing the issues, the issues involved in the conflict should be analysed and it should be understood what this conflict is about. How far it has already involved. Thus, the nature of conflict should be found out. Generally, conflict may arise due to facts, goals, methods, and values. 3. Conflict Handling Modes: There are a number of strategies which we may adopt to resolve the conflict and the important of them are as follows: (a) To avoid appearance of conflict; (b) Not permitting conflict to surface; (c) Mediation; (d) Letting the parties in conflict to settle their scores; and (e) To solve the problems mutually. (	
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Module 5			
	19(a)	Explain the characteristics of Business ethics. Business ethics should take into consideration the following factors: 1. A business should aim to have fair dealing with everyone dealing with it. 2. Ethics should be fixed for everyone working in the organisation at any level and their implementation should be linked with reward- punishment system. 3. Any violation of ethics should be detected at the earliest and remedial measures taken immediately. 4. Business ethics should be based on broad guidelines of what should be done and what should be avoided. 5. The ethics should be based on the perception of what is right.	(7)
	19(b)	Explain the role of computers in technological development. The most commonly discussed cases of computer abuse are instances such as – ☐ The stealing or cheating by employees at work. ☐ The stealing by non-employees or former employees. ☐ The stealing from or cheating clients and consumers. ☐ The violation of contracts for computers sales or services. ☐ The many conspiracies to use computer networks to engage in widespread fraud.	(7)
	20	Explain the advantages and limitations of MNCs. (i) Employment Generation: MNCs create large scale employment opportunities in host countries. This is a big advantage of MNCs for countries; where there is a lot of unemployment. (ii) Automatic Inflow of Foreign Capital: MNCs bring in much needed capital for the rapid development of developing countries. In fact, with the entry of MNCs, inflow of foreign capital is automatic. As a result of the entry of MNCs, India e.g. has attracted foreign investment with several million dollars. (iii) Proper Use of Idle Resources: Because of their advanced technical knowledge, MNCs are in a position to properly utilise idle physical and human resources of the host country. This results in an increase in the National Income of the host country. (iv) Improvement in Balance of Payment Position: MNCs help the host countries to increase their exports. As such, they help the host country to improve upon its Balance of Payment position. (vi) Technical Development:	(14)

	MNCs carry the advantages of technical development to host countries. In fact, MNCs are a vehicle for transference of technical development from one country to another. Because of MNCs poor host countries also begin to develop technically. (i) Danger for Domestic Industries: MNCs, because of their vast economic power, pose a danger to domestic industries; which are still in the process of development. Domestic industries cannot face challenges posed by MNCs. Many domestic industries have to wind up, as a result of threat from MNCs. Thus MNCs give a setback to the economic growth of host countries. (ii) Repatriation of Profits: (Repatriation of profits means sending profits to their country). MNCs earn huge profits. Repatriation of profits by MNCs adversely affects the foreign exchange reserves of the host country; which means that a large amount of foreign exchange goes out of the host country. (iii) No Benefit to Poor People: MNCs produce only those things, which are used by the rich. Therefore, poor people of host countries do not get, generally, any benefit, out of MNCs. Some of the advantages of the MNCs from the viewpoint of the home country are: (i) MNCs usually get raw-materials and labour supplies from host countries at lower prices; specially when host countries are backward or developing economies. (ii) MNCs can widen their market for goods by selling in host countries; and increase their profits. They usually have good earnings by way of dividends earned from operations in host countries. (iii) Through operating in many countries and providing quality services, MNCs add to their international goodwill on which they can capitalize, in the long-run. Some of the limitations of MNCs from the viewpoint of home country may be: (i) There may be loss of employment in the home country, due to spreading manufacturing and marketing operations in other countries. (ii) MNCs face severe problems of managing cultural diversity. This might distract managements' attention from main business issues, causing loss to the home country.	